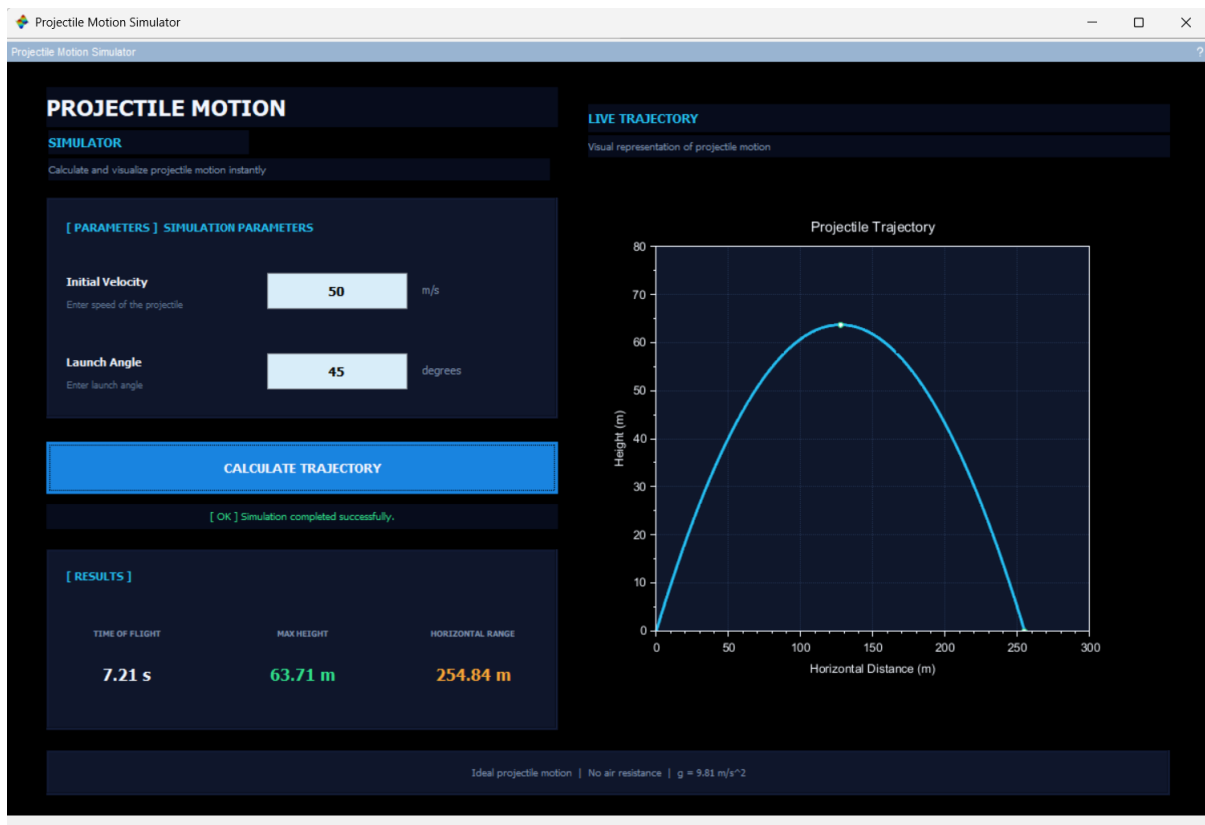


SAMPLE OUTPUTS AND RESULTS

Test Case 1: Standard Launch Parameters

- Input Velocity: 50 m/s
- Input Angle: 45 degrees
- Time of Flight: 7.21 s
- Maximum Height: 63.71 m
- Horizontal Range: 254.84 m
- System Output Message: [OK] Simulation completed successfully.



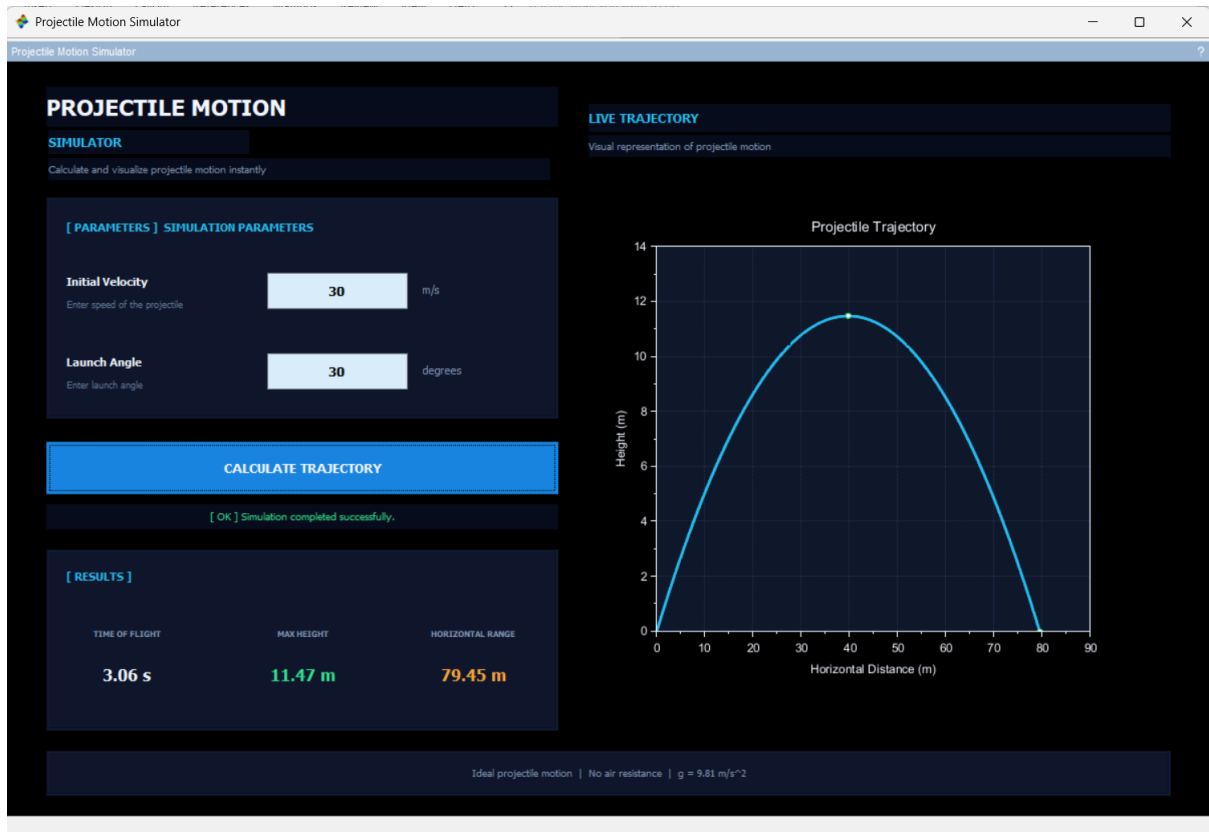
Test Case 2: High Velocity and Steep Angle Launch

- Input Velocity: 80 m/s
- Input Angle: 60 degrees
- Time of Flight: 14.12 s
- Maximum Height: 244.65 m
- Horizontal Range: 564.99 m
- System Output Message: [OK] Simulation completed successfully.



Test Case 3: Low Velocity and Shallow Angle Launch

- Input Velocity: 30 m/s
- Input Angle: 30 degrees
- Time of Flight: 3.06 s
- Maximum Height: 11.47 m
- Horizontal Range: 79.45 m
- System Output Message: [OK] Simulation completed successfully.



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Graphical Trajectory Output Summary:

The application dynamically clears and redraws a 300-point smooth parabolic curve on the right-hand axes panel. Key features of the generated graph include:

- Trajectory Line: Bright cyan line tracing the flight path.
- Peak Apex Marker: Green circular marker highlighted at point (Range / 2, Max Height).
- Impact Point Marker: Green circular marker positioned at landing point (Range, 0).
- Dynamic Axis Bounds: Automatic scaling to fit the trajectory curve with a 10-20% margin above the peak height and maximum distance.